

Common Distributions

Math 742 - Mathematical Statistics

2026-01-16

Overview

This lecture introduces several probability distributions that will serve as canonical examples throughout the course. Our emphasis is on interpretation and structural form, rather than exhaustive derivations.

Several sections are explicitly labeled **READING ONLY**. These sections are included for completeness but will not be covered during lecture.

Discrete Distributions

Bernoulli Distribution (LECTURE CORE)

A Bernoulli random variable models a single trial with two possible outcomes.

If

$$X \sim \text{Bernoulli}(p)$$

then

$$\mathbb{P}(X = 1) = p \text{ and } \mathbb{P}(X = 0) = 1 - p$$

where $0 \leq p \leq 1$.

The Bernoulli distribution is the fundamental building block for many statistical models and will appear repeatedly throughout the course.

Binomial Distribution (LECTURE CORE)

Let

$$X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)(\text{iid})$$

and define

$$Y = \sum_{i=1}^n X_i$$

Then,

$$Y \sim \text{Binomial}(n, p)$$

with probability mass function:

$$\mathbb{P}(Y = y) = \binom{n}{y} p^y (1-p)^{n-y}, y = 0, 1, \dots, n$$

The binomial distribution counts the number of successes in a fixed number of independent trials.

The mean and variance are

$$\mathbb{E}(Y) = np, \quad \mathbb{V}(Y) = np(1-p)$$

Example: Assume that a weighted coin $p = 0.4$ is flipped 5 times. How likely would it be to get at least 4 heads?

Solution: The probability of getting at least 4 heads would equal,

$$\begin{aligned} f(Y = 4) + f(Y = 5) &= \binom{5}{4} (0.4)^4 (1 - 0.4)^1 + \binom{5}{5} (0.4)^5 (1 - 0.4)^0 \\ &= (5)(0.4)^4 (0.6) + (1)(0.4)^5 = 0.08704 \end{aligned}$$

Poisson Distribution (LECTURE CORE)

The Poisson distribution arises as a limit of the binomial distribution when the number of trials is large and the success probability is small.

If $Y \sim \text{Poisson}(\lambda)$ then,

$$\mathbb{P}(Y = y) = \frac{\lambda^y e^{-\lambda}}{y!}; \quad y = 0, 1, 2, \dots$$

where $\lambda > 0$ represents the average rate of occurrence.

The Poisson distribution is commonly used to model counts of events occurring randomly in time or space.

Example: Customers arrive at a department store at a rate of 10 people every 15 minutes. What's the probability that in the next 20 minutes exactly 12 people arrive?

Solution: If the average rate in 15 minutes is 10 people, then the average rate in 20 minutes is $(10/15)(20) \approx 13.33 = \lambda$.

$$\mathbb{P}(y = 12) = \frac{(13.33^{12})e^{-13.33}}{12!} = 0.1068$$

Poisson Distributions Properties (READING ONLY)

$$f_Y(y) = \frac{\lambda^y e^{-\lambda}}{y!}; \quad y = 0, 1, 2, \dots$$

To see that it satisfies the definition of a PMF, we must realize the Taylor series expansion for e^y .

$$e^y = \frac{y^0}{0!} + \frac{y^1}{1!} + \frac{y^2}{2!} + \dots$$

This means that,

$$e^\lambda = \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = \frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \dots$$

Verify that $\sum_{y=0}^{\infty} f_Y(y) = 1$

$$\sum_{y=0}^{\infty} \frac{\lambda^y e^{-\lambda}}{y!} = e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} = e^{-\lambda} e^{\lambda} = 1$$

If Y has a Poisson distribution, what is its mean and variance?

$$\begin{aligned} E(Y) &= \sum_{y=0}^{\infty} y \frac{\lambda^y e^{-\lambda}}{y!}, y = 0, 1, 2, \dots \\ &= 0 \frac{\lambda^0 e^{-\lambda}}{0!} + 1 \frac{\lambda^1 e^{-\lambda}}{1!} + 2 \frac{\lambda^2 e^{-\lambda}}{2!} + \dots + k \frac{\lambda^k e^{-\lambda}}{k!} + \dots \\ &= 0 + \frac{\lambda^1 e^{-\lambda}}{0!} + \frac{\lambda^2 e^{-\lambda}}{1!} + \dots + \frac{\lambda^k e^{-\lambda}}{(k-1)!} + \dots \\ &= \lambda e^{-\lambda} \left(\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \dots + \frac{\lambda^{k-1}}{(k-1)!} + \dots \right) = \lambda e^{-\lambda} e^{\lambda} = \lambda \end{aligned}$$

To show that the variance also equals λ , we first have to determine $\mathbb{E}(Y^2)$. To do this, we rewrite Y^2 as $Y^2 = Y^2 - Y + Y = Y(Y-1) + Y$

$$\mathbb{E}(Y^2) = E[Y(Y-1)] + \mathbb{E}Y = \mathbb{E}[Y(Y-1)] + \lambda$$

$$\begin{aligned} \mathbb{E}[Y(Y-1)] &= \sum_{y=0}^{\infty} y(y-1) \frac{\lambda^y e^{-\lambda}}{y!} \\ &= 0(0-1) \frac{\lambda^0 e^{-\lambda}}{0!} + 1(1-1) \frac{\lambda^1 e^{-\lambda}}{1!} + 2(2-1) \frac{\lambda^2 e^{-\lambda}}{2!} + \dots + k(k-1) \frac{\lambda^k e^{-\lambda}}{k!} + \dots \\ &= 0 + 0 + 2(1) \frac{\lambda^2 e^{-\lambda}}{2!} + 3(2) \frac{\lambda^3 e^{-\lambda}}{3!} + \dots + k(k-1) \frac{\lambda^k e^{-\lambda}}{k!} \\ &= \lambda^2 e^{-\lambda} \left(\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \dots + \frac{\lambda^{k-2}}{(k-2)!} + \dots \right) = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2 \end{aligned}$$

This gives us $\mathbb{E}(Y^2) = \lambda^2 + \lambda$. Therefore,

$$\mathbb{V}(Y) = \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Proof of Poisson PDF

Consider the binomial PDF:

$$f_Y(y) = \binom{n}{y} p^y (1-p)^{n-y}$$

Let $\mathbb{E}(Y) = np = \lambda$, so $p = \lambda/n$

$$\begin{aligned}
f_Y(y) &= \binom{n}{y} \left(\frac{\lambda}{n}\right)^y \left(1 - \frac{\lambda}{n}\right)^{n-y} \\
&= \frac{n!}{(n-y)!y!} \frac{\lambda^y}{n^y} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-y} \\
&= \frac{n!}{(n-y)!n^y} \frac{\lambda^y}{y!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-y} \\
&= \frac{n(n-1)\dots(n-y+1)}{n^y} \frac{\lambda^y}{y!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-y}
\end{aligned}$$

There are y terms in the numerator of the first fraction so,

$$= \left(\frac{n}{n} \frac{n-1}{n} \dots \frac{n-y+1}{n}\right) \frac{\lambda^y}{y!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-y}$$

Now, if n approaches infinity (i.e. take the limit), the first term goes to 1 and we get:

$$\lim_{n \rightarrow \infty} \frac{\lambda^y}{y!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-y}$$

As n goes to infinity, the first term is a constant.

For the middle term, $\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$,

and the last term goes to 1.

Thus,

$$\lim_{n \rightarrow \infty} \frac{\lambda^y}{y!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-y} = \frac{\lambda^y}{y!} e^{-\lambda} (1)$$

$$f_Y(y) = \frac{\lambda^y e^{-\lambda}}{y!}; y = 0, 1, 2, \dots$$

This results in a very good approximation to the binomial distribution with n is large and p is small.

Example: Consider a binomial experiment with $n = 100, p = 0.05$. Let X be the number of successes in n trials. Calculate the binomial probability, $\mathbb{P}(X = 3)$ and then approximate with the Poisson.

$$\text{Exact : } \binom{100}{3} (0.05)^3 (1 - 0.05)^{100-3} = 0.1396$$

Approximate, $\lambda = np = 100(0.05) = 5$

$$\frac{5^3 e^{-5}}{3!} = 0.1404$$

One advantage of Poisson's distribution is that it doesn't require that we know the number of trials, only the average occurrence rate (λ). As an application, think about the number of babies born in a particular hospital. We might only know the average number of births in some amount of time. In order to use Binomial, you need to know the number of trials and the success probability. Poisson only needs the average occurrence rate.

Moment Generating Functions

Sometimes it's easier to derive moments using something called a **moment generating function**.

Definition: The moment generating function of a random variable X is defined as: $M_X(t) = \mathbb{E}[e^{tX}]$

Using the Poisson distribution as an example,

$$M_X(t) = \mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}.$$

The first moment is found by taking the first derivative of the moment generating function:

$$M'_X(t) = \frac{d}{dt} \left[e^{\lambda(e^t - 1)} \right] = e^{\lambda(e^t - 1)} \cdot \lambda e^t$$

(Chain rule)

Now evaluate at $t = 0$,

$$M'_X(0) = e^{\lambda(e^0 - 1)} \cdot \lambda e^0 = \lambda$$

To find the second moment, we evaluate the second derivative at $t = 0$

$$\begin{aligned} M''_X(t) &= \frac{d}{dt} \left[e^{\lambda(e^t - 1)} \cdot \lambda e^t \right] \\ &= e^{\lambda(e^t - 1)} \frac{d}{dt} (\lambda e^t) + \lambda e^t \frac{d}{dt} \left[e^{\lambda(e^t - 1)} \right] \\ &= e^{\lambda(e^t - 1)} \lambda e^t + \lambda e^t e^{\lambda(e^t - 1)} \lambda e^t \\ &= e^{\lambda(e^t - 1)} \lambda e^t (\lambda e^t + 1) \\ M''_X(0) &= e^{\lambda(e^0 - 1)} \lambda e^0 (\lambda e^0 + 1) \\ &= \lambda(\lambda + 1) = \lambda^2 + \lambda \end{aligned}$$

So, $\mathbb{V}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$

Continuous Distributions

Exponential (LECTURE CORE)

The exponential distribution models waiting times between Poisson random events.

If events occur at rate $\lambda > 0$, the waiting time T until the next event has cumulative distribution function

$$F(t) = \begin{cases} 1 - e^{-\lambda t} & : t \geq 0 \\ 0 & : \text{otherwise} \end{cases}$$

Relation to Poisson

Suppose an event just occurred. What is the probability of waiting longer than time t for the next event. The probability of waiting longer than time t is equivalent to the probability of no events in the interval $[0, t]$. Using the Poisson PDF we see that,

$$f(0) = \frac{e^{-\lambda t} \lambda^0}{0!} = e^{-\lambda t}$$

So the probability of waiting less than time t would be,

$$1 - e^{-\lambda t}$$

Memoryless Property (LECTURE CORE)

Let T be a nonnegative continuous random variable representing a waiting time. We say that T has the *memoryless property* if, for all $s, t \geq 0$,

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t).$$

Theorem: If $T \sim \text{Exponential}(\lambda)$ with $\lambda > 0$, then T is memoryless.

Proof: By definition of conditional probability,

$$\mathbb{P}(T > s + t \mid T > s) = \frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)}.$$

For an exponential random variable,

$$\mathbb{P}(T > x) = e^{-\lambda x}, \quad x \geq 0.$$

Therefore,

$$\frac{\mathbb{P}(T > s + t)}{\mathbb{P}(T > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t).$$

This proves that the exponential distribution is memoryless.

Uniqueness

The exponential distribution is the *only* continuous distribution on $[0, \infty)$ with the memoryless property.

Interpretation

The memoryless property implies that the distribution of the remaining waiting time does not depend on how long one has already waited. In particular,

$$\mathbb{P}(T > s + t \mid T > s) = \mathbb{P}(T > t)$$

means that, given survival up to time s , the additional waiting time has the same distribution as the original waiting time.

Connection to Poisson Processes

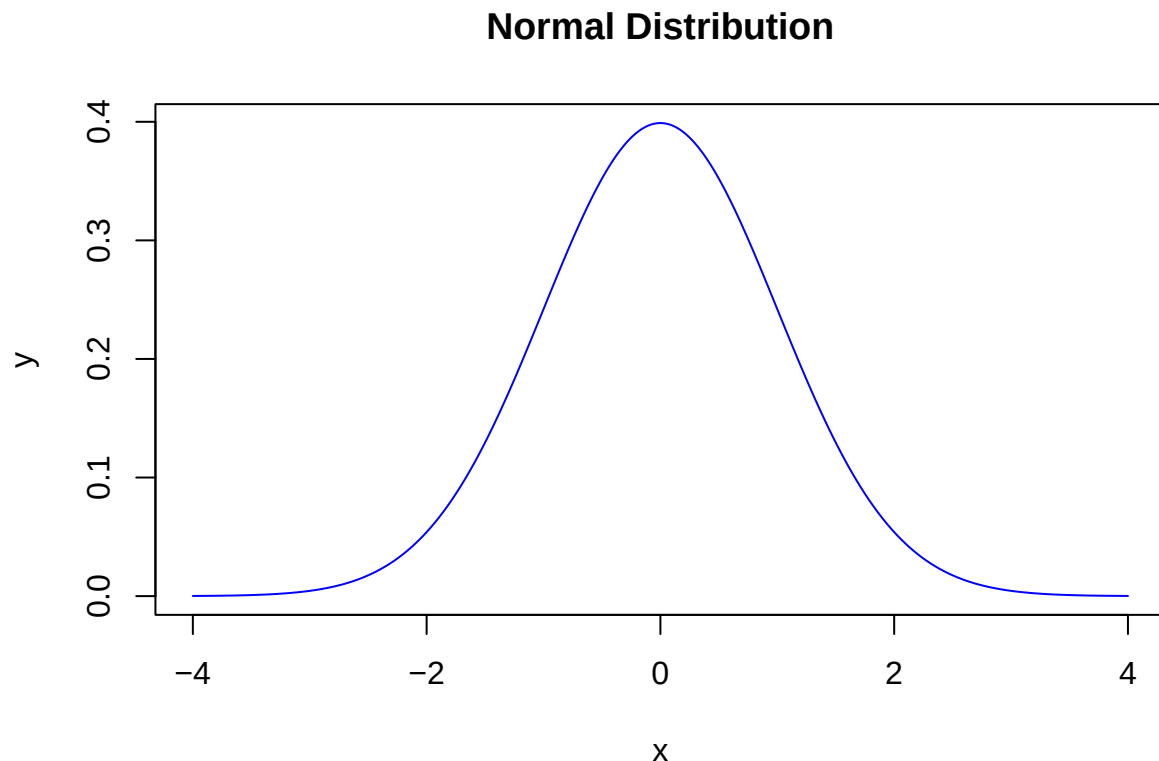
If events occur according to a Poisson process with rate λ , then the waiting time between successive events is exponentially distributed with parameter λ . The memoryless property reflects the lack of “aging” in the Poisson process: the future evolution of the process is independent of the past.

Remark

In contrast, most continuous distributions (e.g., Weibull, Gamma) do not satisfy the memoryless property; in such cases, the probability of waiting longer typically depends on how much time has already elapsed.

Normal Distribution (LECTURE CORE)

The normal distribution is the proverbial “bell-shaped” curve.



The **Normal Distribution** has 2 parameters, μ and σ^2 which are the mean and variance. The PDF for the normal distribution has the following equation:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, -\infty < x < \infty$$

If $X \sim N(\mu, \sigma^2)$ and $Z = \frac{x-\mu}{\sigma}$ then Z has a standard normal distribution. A **standard normal** distribution has $\mu = 0, \sigma = 1$.

Why?

$$\begin{aligned} \mathbb{P}(Z \leq z) &= \mathbb{P}\left(\frac{X - \mu}{\sigma} \leq z\right) \\ &= \mathbb{P}(X \leq z\sigma + \mu) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{z\sigma+\mu} e^{-(x-\mu)^2/(2\sigma^2)} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt, \text{ where } t = \frac{x-\mu}{\sigma}
\end{aligned}$$

This proves that $\mathbb{P}(Z \leq z)$ is the standard normal CDF.

$$\mathbb{E}(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2} dz = \frac{-1}{\sqrt{2\pi}} e^{-z^2/2} \Big|_{-\infty}^{\infty} = 0$$

$$\mathbb{E}(X) = \mathbb{E}(\mu + \sigma Z) = \mu + \sigma \mathbb{E}(Z) = \mu$$

To show that $f_Z(z)$ is a legitimate PDF we have to integrate it over all values of z and show that it equals 1.

We wish to evaluate,

$$\int_{-\infty}^{\infty} e^{-z^2/2} dz.$$

Our hope is that this integral equals $\sqrt{2\pi}$.

Since the integrand is an even function,

$$\int_{-\infty}^{\infty} e^{-z^2/2} dz = 2 \int_0^{\infty} e^{-z^2/2} dz.$$

Let

$$I = \int_0^{\infty} e^{-z^2/2} dz.$$

Then

$$I^2 = \left(\int_0^{\infty} e^{-t^2/2} dt \right) \left(\int_0^{\infty} e^{-u^2/2} du \right) = \int_0^{\infty} \int_0^{\infty} e^{-(t^2+u^2)/2} dt du,$$

Convert to polar coordinates by letting

$$t = r \cos \theta, \quad u = r \sin \theta,$$

with Jacobian determinant r . Since $t, u > 0$, the region of integration is the first quadrant, corresponding to

$$0 < r < \infty, \quad 0 < \theta < \frac{\pi}{2}.$$

Thus,

$$I^2 = \int_0^{\pi/2} \int_0^{\infty} e^{-r^2/2} r dr d\theta.$$

Integrating with respect to θ ,

$$I^2 = \frac{\pi}{2} \int_0^\infty r e^{-r^2/2} dr.$$

Now substitute $u = r^2/2$, so that $du = r dr$:

$$\int_0^\infty r e^{-r^2/2} dr = \int_0^\infty e^{-u} du = 1.$$

Therefore,

$$I^2 = \frac{\pi}{2}, \quad I = \sqrt{\frac{\pi}{2}}.$$

Finally,

$$\int_{-\infty}^\infty e^{-z^2/2} dz = 2I = \sqrt{2\pi}.$$

We find inflection points by solving $f_Z''(z) = 0$ and checking sign change.

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$\frac{d}{dz} \left[\frac{1}{\sqrt{2\pi}} e^{-z^2/2} \right] = \frac{-z}{\sqrt{2\pi}} e^{-z^2/2} = -z f_Z(z)$$

Taking another derivative,

$$\frac{d}{dz} (-z f_Z(z)) = -z f_Z'(z) + f_Z(z)(-1) = -z f_Z'(z) - f_Z(z)$$

Now, substitute $f_Z'(z) = -z f_Z(z)$ so,

$$= -z(-z f_Z(z)) - f_Z(z) = z^2 f_Z(z) - f_Z(z) = (z^2 - 1) f_Z(z)$$

Set the second derivative equal to zero to find the inflection points.

$$(z^2 - 1) f_Z(z) = 0$$

$$z^2 - 1 = 0$$

$$z = \pm 1$$

1 and minus 1 are candidates for inflection points. We need to check concavity.

It's pretty easy to show that $z = -2$ the second derivative is positive, at $z = -1/2$ the second derivative is negative so $z = -1$ is an inflection point. Likewise, at $z = 1/2$ the second derivative is negative and at $z = 2$ the second derivative is positive so $z = 1$ is an inflection point.

Summary of Core Distributions

Distribution	PDF	Mean	Variance
Binomial(n, p)	$\binom{n}{x} p^x (1-p)^{(n-x)}$	np	$np(1-p)$
Poisson(λ)	$\frac{(\lambda)^x e^{-\lambda}}{x!}$	λ	λ
Exponential	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	$\frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

Connection to Exponential Families and Sufficiency

The distributions introduced in here share a common mathematical structure. Next it will be shown that the Bernoulli, Binomial, Poisson, Exponential, and Normal distributions all belong to the **exponential family**.

This structure explains the existence of sufficient statistics and simplifies likelihood-based inference.

Additional Distributions (READING ONLY)

Multinomial Distribution

The multivariate version of the binomial distribution is called the **Multinomial** distribution. Suppose poker chips are placed in a bag. The poker chips can have different colors, we'll label them, color 1, color 2, ..., color k . Let p_1 be the probability of drawing color 1, p_2 be the probability of drawing color 2, ..., p_k is the probability of drawing color k ($p = (p_1, p_2, \dots, p_k)$). Draw from the bag n times with replacement. The random variable X_j is the number color j is drawn. $X = (X_1, X_2, \dots, X_k)$ and $n = \sum_{j=1}^n X_j$. Then, X has a multinomial distribution with,

$$f_X(x) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

where, $\binom{n}{x_1, x_2, \dots, x_k} = \frac{n!}{x_1! x_2! \dots x_k!}$

Example: Suppose that 7 faculty members at SD Mines are chosen for a committee. The departments that will offer up a professor are Civil (11 faculty members), Mechanical (8 faculty members), Humanities (10 faculty members) and Math (9 faculty members). Determine the probability that the committee consists of 1 Civil, 2 Mechanical, 3 Humanities and 1 Math.

Let $n = 11 + 8 + 10 + 9 = 38$, $X = (1, 2, 3, 1)$, p_1 (Civil) = $11/38$, p_2 (ME) = $8/38$, p_3 (Humanities) = $10/38$ and p_4 (Math) = $9/38$

$$X_1 + X_2 + X_3 + X_4 = 7.$$

$$\mathbb{P}(X_1 = 1, X_2 = 2, X_3 = 3, X_4 = 1) = \frac{7!}{1!2!3!1!} (11/38)^1 (8/38)^2 (10/38)^3 (9/38)^1 \approx 0.0233$$

Multivariate Normal Distribution

The multivariate normal extends the univariate case to correlated random variables. It is central to multivariate regression, PCA, and Gaussian processes.

Let

$$Z = \begin{pmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_k \end{pmatrix},$$

where $Z_1, Z_2, \dots, Z_k$ are independent standard normal random variables. The PDF for Z is thus,

$$f_Z(z) = \prod_{i=1}^k f_Z(Z_i) = \frac{1}{(2\pi)^{k/2}} e^{-\frac{1}{2} \sum_{j=1}^k z_j^2} = \frac{1}{(2\pi)^{k/2}} e^{-\frac{1}{2} z^T z}$$

This is called a standard multivariate normal distribution ($N(0, I)$).

$$\mu = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

and μ has length k .

and I is a $k \times k$ identity matrix.

The more general multivariate normal distribution would be $X \sim N(\mu, \Sigma)$ whose PDF is:

$$f(x; \mu, \Sigma) = \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)}$$

where $|\Sigma|$ is the determinate of Σ , μ is a column vector of means (length = k), Σ is a $k \times k$ positive definite matrix.

Note: a positive definite matrix is a matrix such that for all nonzero vectors x , $x^T \Sigma x > 0$.

Σ is known as the variance covariance matrix. For $X = (X_1, X_2, \dots, X_k)^T$,

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \cdots & \text{Var}(X_p) \end{bmatrix}$$